

Proposition 3.1. *Let $d = d_1 + d_2$, $\mathbf{u} = \mathbf{u}_1 \oplus \mathbf{u}_2 \in \mathbb{R}^{2d_1} \oplus \mathbb{R}^{2d_2}$ and $L \in Sp(2d, J)$ with $L^{-1} = \begin{bmatrix} L_{11} & L_{12} \\ L_{21} & L_{22} \end{bmatrix}$, $L_{jj} \in M_{2d_j}(\mathbb{R})$, $j = 1, 2$. Then the map $\Psi : \mathcal{T}(\Gamma(\mathbb{C}^{d_1})) \rightarrow \mathcal{T}(\Gamma(\mathbb{C}^{d_1}))$ defined by*

$$\Psi(\varrho) := \text{tr}_2 \left(W(\mathbf{u})\Gamma(L)(\varrho \otimes \rho_{(0, I_{d_2})})(W(\mathbf{u})\Gamma(L))^\dagger \right), \quad \forall \varrho \in \mathcal{T}(\Gamma_s(\mathbb{C}^{d_1})), \quad (6)$$

is a quantum channel that maps Gaussian states to Gaussian states. Indeed,

$$\Psi(\rho_{(\mathbf{m}, S)}) = \rho_{(L_{11}^T \mathbf{m} + 2J_{2d_1} \mathbf{u}_1, L_{11}^T S L_{11} + L_{21}^T L_{21})}, \quad (7)$$

for every Gaussian states $\rho_{(\mathbf{m}, S)}$ on $\Gamma(\mathbb{C}^{d_1})$.

Proof. Since $U := W(\mathbf{u})\Gamma(L)$ is a unitary, by Stinespring representation theorem, we know that Ψ is a quantum channel. Further, given any Gaussian state $\rho_{(\mathbf{m}, S)}$ the quantum characteristic function of $\Psi(\rho_{(\mathbf{m}, S)})$ at any $\mathbf{v} \in \mathbb{R}^{2d_1}$ is given by

$$\begin{aligned} \Psi(\widehat{\rho_{(\mathbf{m}, S)}})(\mathbf{v}) &= \text{tr} \left(\text{tr}_2 \left(W(\mathbf{u})\Gamma(L)(\rho_{(\mathbf{m}, S)} \otimes \rho_{(0, I_{d_2})})(W(\mathbf{u})\Gamma(L))^\dagger \right) W(\mathbf{v}) \right) \\ &= \text{tr} \left((\rho_{(\mathbf{m}, S)} \otimes \rho_{(0, I_{d_2})}) \Gamma(L)^\dagger W(-\mathbf{u}) W(\mathbf{v} \oplus \mathbf{0}) W(\mathbf{u}) \Gamma(L) \right) \\ &= \text{tr} \left((\rho_{(\mathbf{m}, S)} \otimes \rho_{(0, I_{d_2})}) \Gamma(L^{-1}) W(\mathbf{v} \oplus \mathbf{0}) \Gamma(L^{-1})^\dagger \right) \exp\{2i \text{Im} \langle \mathbf{u}_1 | \mathbf{v} \rangle\} \\ &= \text{tr} \left((\rho_{(\mathbf{m}, S)} \otimes \rho_{(0, I_{d_2})}) W(L^{-1}(\mathbf{v} \oplus \mathbf{0})) \right) \exp\{2i \mathbf{u}_1^T J_{2d_1} \mathbf{v}\} \\ &= \text{tr} \left((\rho_{(\mathbf{m}, S)} \otimes \rho_{(0, I_{d_2})}) W(L_{11} \mathbf{v} \oplus L_{21} \mathbf{v}) \right) \exp\{2i \mathbf{u}_1^T J_{2d_1} \mathbf{v}\} \\ &= \text{tr}(\rho_{(\mathbf{m}, S)} W(L_{11} \mathbf{v})) \text{tr}(\rho_{(0, I_{d_2})} W(L_{21} \mathbf{v})) \exp\{2i \mathbf{u}_1^T J_{2d_1} \mathbf{v}\} \\ &= \exp \left\{ -i (L_{11}^T \mathbf{m} + 2J_{2d_1} \mathbf{u}_1)^T \mathbf{v} - \frac{1}{2} \mathbf{v}^T (L_{11}^T S L_{11} + L_{21}^T L_{21}) \mathbf{v} \right\}. \end{aligned}$$

Since $\mathbf{v} \in \mathbb{R}^{2d_1}$ is arbitrary we conclude that $\Psi(\rho_{(\mathbf{m}, S)})$ is a Gaussian state with mean $L_{11}^T \mathbf{m} + 2J_{2d_1} \mathbf{u}_1$ and covariance $L_{11}^T S L_{11} + L_{21}^T L_{21}$. $\square$

To achieve our main goal, we will first establish some technical results.

Lemma 3.1. *Let $\lambda_j \in [-1, 1]$ for $1 \leq j \leq d$. Then there exists $Q \in M_{2d \times 4d}(\mathbb{R})$ such that $QQ^T = I_{2d}$ and*

$$Q \left(\bigoplus_{j=1}^{2d} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \right) Q^T = \bigoplus_{j=1}^d \begin{bmatrix} 0 & \lambda_j \\ -\lambda_j & 0 \end{bmatrix}.$$

Proof. For any $\theta \in \mathbb{R}$, let

$$Q(\theta) := \begin{bmatrix} \cos \theta & 0 & -\sin \theta & 0 \\ 0 & \cos \theta & 0 & \sin \theta \end{bmatrix}.$$

Then, we have $Q(\theta)Q(\theta)^T = I_2$, and

$$Q(\theta) \left(\bigoplus_{j=1}^2 \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \right) Q(\theta)^T = \begin{bmatrix} 0 & \cos 2\theta \\ -\cos 2\theta & 0 \end{bmatrix}.$$

Now, for each $1 \leq j \leq d$, choose and fix $\theta_j \in \mathbb{R}$ be such that $\cos 2\theta_j = \lambda_j$. Then $Q := \bigoplus_{j=1}^d Q(\theta_j)$ gives the required matrix. $\square$

Corollary 3.1. *Let $A \in M_d(\mathbb{R})$ be a positive contraction. Then there exists a matrix $Q \in M_{2d \times 4d}(\mathbb{R})$ such that $QQ^T = I_{2d}$ and $QJ_{4d}Q^T = \begin{bmatrix} 0 & A \\ -A & 0 \end{bmatrix}$.*